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Date:

A.

Bayes' rule: $P(Y|X) = \frac{P(X|Y) \cdot P(Y)}{P(X)}$

$P(Y)$: prior probability of Y before observing X

$P(X|Y)$: likely hood of class X given class Y

$P(Y|X)$: posterior probability of class Y after class X.

$P(X)$: marginal probability of observing X.

B. Naive Bayes assume all features $X_1, X_2, \ldots, X_n$ are conditionally ~~indeep~~ independant given class Y. This means once the class is known, knowing 1 features provde no additional info about another. $P(x_1, \ldots x_n|y) = \Pi_{i=1}^{n} P(X_i|Y)$

C. Smoothing prevent 0probabilites when a feature value has never appeared in the training data for a class. Without smoothing, a 0 conditional probability can make the product of feature likelihood $= 0 \rightarrow$ class has 0 posteror probability. For categorical features, add-one smoothing adds one to each observed count and adjust denominator accordingly -> allow unseen feature values to receive nonzero probability.

TEKO SOLUTION

Date: ____

D. NB is generative because it uses $P(Y)$ and $P(X|Y)$ to represent $P(X,Y) = P(X|Y)\,P(Y)$, then apply Bayes rule to calculate $P(Y|X)$. Discriminative model like logistic regression directly model $P(Y|X)$ without explicitly modeling how features are generated.

E.

$$P(Y=1|X=1) = \frac{P(X=1|Y=1)\,P(Y=1)}{P(X=1)}$$

$$P(X=1) = 0,8.(0,3) + 0,2.(0,7) = 0,38$$

$$P(Y=1|X=1) = \frac{0,8\,(0,3)}{0,38} \simeq 0,6316$$

→ Posterior probability is 63,16 %